

# Tahsin Reza

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## EDUCATION

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**Bangladesh University of Engineering and Technology (BUET), Dhaka**  
Bachelor of Urban and Regional Planning (BURP)

Jan 2022 - June 2026

**CGPA:** 3.70/4.00

**Thesis:** Flow Characteristics of Traditional Rickshaws and Battery-Operated Three-Wheelers in Non-Lane-Based Heterogeneous Traffic of Dhaka

**Relevant Coursework:** Statistics I and II, Programming Techniques, GIS and Remote Sensing, Transportation Policy and Planning, Environmental Planning and Urban Planning Techniques

## RESEARCH INTEREST

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Travel Behavior & Mode Choice, Geospatial Analytics for Transportation, Data Science for Transport Planning, Urban Mobility, Safety, Accessibility & Equity in Transportation

## ACADEMIC HONORS AND AWARDS

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- University Merit Scholarship, BUET (2023)
- University Stipend, BUET (2022)

## RESEARCH EXPERIENCE

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### [1] Time-Headway Analysis of Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic

*Shared vehicle-tracking pipeline, Tikatuli and Shahjahanpur corridors, Dhaka*

- Built an automated extraction pipeline, a YOLO26 detector trained on roughly 207,000 self-labeled vehicle instances paired with BoT-SORT tracking, producing about 900 follower-leader observations across two corridors that feed both manuscripts below.

#### (a) Pair-Specific Time Headway Distributions and Covariate-Conditioned AFT Models for Pedal Rickshaws and Battery-Operated Three-Wheelers in Dhaka's Non-Lane-Based Heterogeneous Traffic

*Manuscript Submitted (Transportation Research Board (TRB) Annual Meeting 2027, 1st Author)* [\[Pre-print Version\]](#)

- Fitted pair-specific headway distributions and accelerated failure time (AFT) models to explain distributional characteristics and shifts, accounting for covariate effects that feed both simulation modeling and the policy horizon.

#### (b) One Label, Two Vehicles: Time-Headway Evidence Against Treating Pedal and Battery Rickshaws Alike in Mixed Traffic

*Manuscript in Preparation*

- Modeled time headway as a function of spacing and speed using a maximum-likelihood location-scale regression to isolate which of the two drives the class gap rather than assuming it; found battery three-wheelers close 66-80% of their shorter following margin through higher speed alone, exposing them to critically short headways at 2-4x the rate of pedal rickshaws, replicated across two independent corridors.

### [2] Monomodal versus Intermodal Trip Choice in Dhaka: Bayesian Multilevel Models and Explainable Machine Learning

*Manuscript in Preparation*

- Built a four-level (M0–M3) Bayesian multilevel logistic regression and an explainable Random Forest/SHAP pipeline to model monomodal vs. intermodal trip choice across 25,242 trips nested within 3,729 Dhaka households.
- Identified household vehicle ownership as the strongest driver of trip modality and rickshaw-plus-walking as the dominant intermodal chain (55% of 6,838 intermodal trips).

### [3] Flow Characteristics of Traditional Rickshaws and Battery-Operated Three-Wheelers in Non-Lane-Based Heterogeneous Traffic of Dhaka

*Bachelor's Thesis, Department of Urban and Regional Planning, BUET (2026)* [\[Thesis Report\]](#)

- Conducted videographic surveys at three Dhaka roads, manually extracting 531 valid target-leader time-headway observations using a reference-line method.

- Benchmarked six tree-based models and selected a tuned CatBoost model (Test  $R^2 = 0.674$ , MAE = 0.400 s) with SHAP analysis identifying speed as the strongest predictor, confirming battery-operated three-wheelers consistently follow with shorter headways than rickshaws - evidence against regulating them as one vehicle class.

## SELECTED UNDERGRADUATE PROJECTS

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### [1] Analyzing Core Rural Road Networks and Land Use Conversion: A Case Study of Selected Upazilas in Bangladesh

Rural Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed all unions across 14 upazilas via Key Informant Interviews and combined this with GIS buffer analysis of LGED road, BBS demographic, and Sentinel land-use data, then built a Multiple Linear Regression model (log-log form,  $R^2 = 0.44$ ,  $F = 24.25$ ,  $p < 0.001$ ) to identify the drivers of core road network density.
- Found population density ( $\beta = 0.305$ ) and educational institution density ( $\beta = 0.205$ ) to be the strongest predictors: core roads served roughly 90% of growth centers and village markets and 50–60% of population, education, and healthcare facilities per upazila, while natural land-use conversion occurred in 81% of unions.

### [2] A Study on Factors Influencing Access Mode Choice of Metro Rail (MRT) Users in Dhaka City

Social and Physical Survey Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 345 MRT users across three stations (Agargaon, Uttara North, and Pallabi) via random sampling to identify factors influencing first- and last-mile (FLM) access mode choice
- Found rickshaw use dominant overall (41%), with walking and bus tied at 20% each; over 84% of users traveled from within 5 km and about 77% of trips under 1 km were made on foot, and mode choice varied by station, walking led in Pallabi (shortest average distance, 1.39 km), rickshaw use was highest in Uttara North (lower perceived crime risk), and bus use was steadiest in Agargaon (more consistent road width).

### [3] Community-Based Assessment of Institutional Capacity and Public Awareness for Urban Earthquake Risk Reduction: A Case Study of Ward 53, Dhaka

Participatory Planning Studio, Department of Urban and Regional Planning, BUET [\[Presentation\]](#)

- Applied Participatory Capacity and Vulnerability Analysis (PCVA) - vulnerability mapping, timeline analysis, cause-effect diagrams, process mapping, and Venn diagrams, through focus group discussions and key informant interviews in Ward 53, Dhaka South City Corporation.
- Documented acute physical vulnerability (roads averaging 2.2 m wide serving five- to six-story buildings on un-setback ~720 sq. ft. plots) and, via pairwise ranking and cause-effect analysis, traced structural risk to fear-based concealment of unsafe buildings and unenforced BNBC approval procedures.

### [4] Factors Influencing Pedestrian Catchment Area of Major MRT Stations in Dhaka

Transport Planning Studio, Department of Urban and Regional Planning, BUET [\[Report\]](#)

- Surveyed 717 commuters (401 at Motijheel, 316 at Farmgate) via random sampling and station-gate video counts, mapped built-environment variables across six GIS buffer rings (250–2000 m), and trained a Random Forest Regressor (scikit-learn) to predict walking distance.
- Found mean egress-walk distances of 440.8 m (Motijheel) and 401.7 m (Farmgate), with 12–17% of trips extending beyond 1 km (well past the conventional 800 m catchment radius) while the model reached  $R^2 = 0.94$ , with road density and floor area outweighing individual factors like age or gender.

### [5] Mughal Nurture: Heritage Conservation and Tourism Planning for Old Dhaka [\[Report\]](#)

Urban Planning Studio, Department of Urban and Regional Planning, BUET

- Surveyed structural condition and surrounding land use at five Mughal-era heritage sites in Old Dhaka and developed site-specific conservation and tourism proposals addressing authenticity, height restriction, streetscape design, and traffic management for each site.

### [6] Transportation Problem Solver: North-West Corner and Least Cost Methods [\[Presentation\]](#)

Programming Techniques, Department of Urban and Regional Planning, BUET

- Built a Python tool that computes and compares initial feasible solutions to the classical transportation problem using both the North-West Corner and Least Cost methods.

## PROFESSIONAL EXPERIENCE

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### Assistant Urban Planner

July 2026 - Present

Sheltech Consultants (Pvt.) Ltd., Dhaka

- Developed the transport planning part of the report of a Pourashava master plan project from scratch, newly introducing capacity, flow, level-of-service, and Space Syntax analysis into the plan.
- Support the wider team on transport-related planning questions across the project.

## Intern

November 2025 - December 2025

Dhaka Transport Coordination Authority (DTCA), Dhaka

- Assessed the traffic impact of converting a road in Dhaka into a pedestrian precinct. Ran traffic and pedestrian surveys, then built and compared before-and-after microsimulation scenarios in PTV Vissim to test the proposed layout.

## SOFTWARE AND PROGRAMMING SKILLS

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**Programming Language:** Python, R, C | **GIS Tool:** ArcGIS, ArcGIS Pro | **Transportation Modelling:** PTV Vissim | **Modeling and ML:** Discrete Choice Models, Survival/AFT Models, SHAP, YOLO Object Detection | **Design and Drafting:** AutoCAD, Adobe Illustrator, Adobe Photoshop | **Productivity:** MS Word, Excel, PowerPoint, Project, CapCut

## EXTRA-CURRICULAR ACTIVITIES

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- Secretary, Student Chapter, **Bangladesh Institute of Planners (BIP)** (2025-2026)
- Vice President, **URP Students' Association of BUET (USAB)** (2025-2026)
- Joint General Secretary, **Help\_BUETian** (2025-2026)
- Excel Skills for Business Specialization:** Coursera certification in Excel-based data analysis, forecasting and dashboarding (2023)
- Tutoring** Secondary and Higher Secondary students (Physics, Math, ICT, Chemistry) (2021-Present)

## REFERENCES

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### Dr. Md. Musleh Uddin Hasan

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Dept. of Urban and Regional Planning  
Bangladesh University of Engineering and Technology  
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Relationship: Thesis Supervisor and Research Mentor

### Ms. Sadia Afroj

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Relationship: Course Instructor and Research Mentor

### Dr. Md. Asif Raihan

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